Name:	Student ID:

CONVEXITY AND SVMS PROBLEM SHEET

This sheet covers convexity, support vector machines and kernels. Questions 1 and 2 are about convex functions and Jensen's inequality. Question 3 solves a tiny SVM by hand, in both the primal and the dual form. Question 4 asks which functions are valid kernels.

1

(a) Starting from the definition of a convex function

$$f(ax + (1-a)y) \leq af(x) + (1-a)f(y) \tag{1}$$

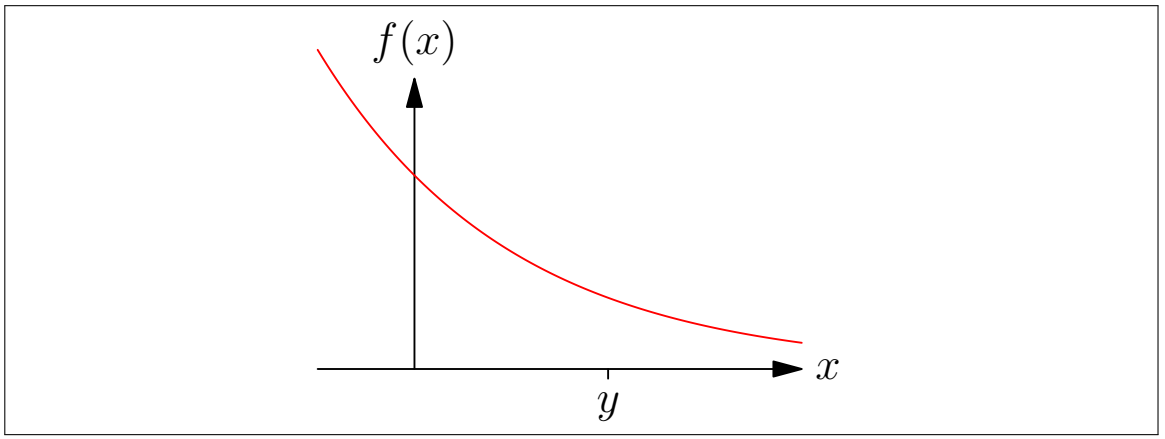
Let $a = \epsilon/(x-y)$ and rearrange the inequality to give

$$(x-y)\left(\frac{f(y+\epsilon) - f(y)}{\epsilon}\right)$$

on the left-hand side. Taking the limit $\epsilon \to 0$ show that the function $f(x)$ lies above the tangent line $t(x) = f(y) + (x-y)f'(y)$ going through the point y. [4 marks]

4

(b) Sketch the tangent line, $t(x)$, at the point y in the graph shown below. [1 mark]



$\frac{}{1}$

(c) Starting from the inequality for a convex function

$$f(x) \geq f(y) + (x - y) f'(y) \qquad (2)$$

consider the case $y = x + \epsilon$, then by Taylor expanding $f(x + \epsilon)$ and $f'(x + \epsilon)$ around x and keeping all terms up to order ϵ^2 show that for a convex function $f''(x) \geq 0$. [4 marks]

$\frac{}{4}$

(d) Prove that $-\log(x)$ is convex-up for $x > 0$. [1 mark]

$\frac{}{1}$

End of question 1

(a) $\frac{}{4}$	(b) $\frac{}{1}$	(c) $\frac{}{4}$	(d) $\frac{}{1}$	Total $\frac{}{10}$

2

(a) If $\|\boldsymbol{x}\|$ is a proper norm use the triangular inequality ($\|\boldsymbol{x}+\boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$), "linearity" of a norm ($\|a\boldsymbol{x}\| = |a|\,\|\boldsymbol{x}\|$) and the definition of convexity, to show that the norm is convex. [5 marks]

5

(b) Consider a classification problem where $\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})$ is the probability that a learning machine with parameters $\boldsymbol{\theta}$ predicts that input $\boldsymbol{x}$ belongs to class $c \in \mathcal{C}$. Assume the training is stochastic so the probability of obtaining parameters $\boldsymbol{\theta}$ is $\rho(\boldsymbol{\theta})$. Let $\hat{m}_c(\boldsymbol{x}) = \mathbb{E}_{\boldsymbol{\theta}}\Big[\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})\Big]$ be the output of the mean machine for class c. Assuming that for a data point $(\boldsymbol{x},y)$, where y is a class label, we use a cross entropy loss

$$L(\boldsymbol{x},y,\boldsymbol{\theta}) = -\sum_{c\in\mathcal{C}} [\![y=c]\!] \log\Big(\hat{f}_c(\boldsymbol{x}|\boldsymbol{\theta})\Big),$$

show that the expected loss over inputs and parameters can be written as the expected loss of the mean machine plus a second loss. Use Jensen's inequality ($\mathbb{E}\big[\log(X)\big] \leq \log\big(\mathbb{E}\big[X\big]\big)$) to show the second term is non-negative. [5 marks]

5

End of question 2

(a) $\frac{}{5}$	(b) $\frac{}{5}$	Total $\frac{}{10}$

3 In the lectures the maximum-margin SVM finds $\boldsymbol{w}$ and b that minimise $\frac{1}{2}\|\boldsymbol{w}\|^2$ subject to $y_k(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_k - b) \geq 1$ for every data point. The dual problem is to maximise

$$\mathcal{L}(\boldsymbol{\alpha}) = \sum_k \alpha_k - \frac{1}{2}\sum_{k,l} \alpha_k \alpha_l y_k y_l \boldsymbol{x}_k^\mathsf{T} \boldsymbol{x}_l$$

subject to $\sum_k \alpha_k y_k = 0$ and $\alpha_k \geq 0$, after which $\boldsymbol{w} = \sum_k \alpha_k y_k \boldsymbol{x}_k$. In this question we solve a tiny problem by hand.

(a) Take two points, $\boldsymbol{x}_1 = (1,1)^\mathsf{T}$ with $y_1 = +1$ and $\boldsymbol{x}_2 = (-1,-1)^\mathsf{T}$ with $y_2 = -1$. By symmetry the solution has $b = 0$ and $\boldsymbol{w} = (u,u)^\mathsf{T}$. Find u and the margin $1/\|\boldsymbol{w}\|$. Check that the margin is half the distance between the two points. [2 marks]

2

(b) Now solve the dual problem for the same two points. Show that the constraint forces $\alpha_1 = \alpha_2 = \alpha$, write $\mathcal{L}$ as a function of α and maximise it. Recover $\boldsymbol{w}$ and check that it agrees with part (a). Find b from the fact that $\boldsymbol{x}_1$ lies on the margin, $y_1(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_1 - b) = 1$. [3 marks]

3

(c) Add a third point, $\boldsymbol{x}_3 = (3,0)^\mathsf{T}$ with $y_3 = +1$. Without solving the problem again, explain why the solution does not change, and why $\alpha_3 = 0$. [2 marks]

$\overline{2}$

(d) Classify the new point $\boldsymbol{x} = (-2,1)^\mathsf{T}$ using the dual form $\operatorname{sgn}\big(\sum_k \alpha_k y_k \boldsymbol{x}_k^\mathsf{T}\boldsymbol{x} - b\big)$. [1 mark]

$\overline{1}$

(e) Return to the two points of part (a), but use a soft margin with $C = 0.1$, so that the dual constraint becomes $0 \le \alpha_k \le C$. Find α, $\boldsymbol{w}$ and the margin. Taking $b = 0$, find the slack $s_k = 1 - y_k(\boldsymbol{w}^\mathsf{T}\boldsymbol{x}_k - b)$ of each point. [2 marks]

$\overline{2}$

End of question 3

(a) $\overline{2}$	(b) $\overline{3}$	(c) $\overline{2}$	(d) $\overline{1}$	(e) $\overline{2}$	Total $\overline{10}$

4 A function $K(\boldsymbol{x},\boldsymbol{y})$ is a valid (positive semi-definite) kernel if it can be written as $K(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{\phi}(\boldsymbol{x})^\mathsf{T}\boldsymbol{\phi}(\boldsymbol{y})$ for some real feature vector $\boldsymbol{\phi}(\boldsymbol{x})$. Equivalently, for any points $\boldsymbol{x}_1,\ldots,\boldsymbol{x}_n$ the Gram matrix with elements $K_{ij} = K(\boldsymbol{x}_i,\boldsymbol{x}_j)$ is positive semi-definite.

(a) For $\boldsymbol{x},\boldsymbol{y} \in \mathbb{R}^2$, find a feature vector $\boldsymbol{\phi}(\boldsymbol{x})$ with six components such that $(1 + \boldsymbol{x}^\mathsf{T}\boldsymbol{y})^2 = \boldsymbol{\phi}(\boldsymbol{x})^\mathsf{T}\boldsymbol{\phi}(\boldsymbol{y})$. [2 marks]

2

(b) Write down the Gram matrix of the linear kernel $K(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{x}^\mathsf{T}\boldsymbol{y}$ for the three points $\boldsymbol{x}_1 = (1,0)^\mathsf{T}$, $\boldsymbol{x}_2 = (0,1)^\mathsf{T}$ and $\boldsymbol{x}_3 = (1,1)^\mathsf{T}$, and find its eigenvalues. Why is one of them zero? [2 marks]

2

(c) Show that every valid kernel has $K(\boldsymbol{x},\boldsymbol{x}) \geq 0$. Hence show that $K(\boldsymbol{x},\boldsymbol{y}) = \boldsymbol{x}^\mathsf{T}\boldsymbol{y} - 1$ is not a valid kernel. [2 marks]

2

(d) Show that $K(\boldsymbol{x},\boldsymbol{y}) = \|\boldsymbol{x} - \boldsymbol{y}\|^2$ is not a valid kernel, even though $K(\boldsymbol{x},\boldsymbol{x}) \geq 0$. (Hint: look at the Gram matrix of two different points.) [2 marks]

$\frac{}{2}$

(e) Using the rules from the lecture (sums, positive multiples, products and exponentials of valid kernels are valid), show that $K(\boldsymbol{x},\boldsymbol{y}) = 3(\boldsymbol{x}^\mathsf{T}\boldsymbol{y})^2 + \exp(\boldsymbol{x}^\mathsf{T}\boldsymbol{y})$ is a valid kernel. [2 marks]

$\frac{}{2}$

End of question 4

(a) $\frac{}{2}$	(b) $\frac{}{2}$	(c) $\frac{}{2}$	(d) $\frac{}{2}$	(e) $\frac{}{2}$	Total $\frac{}{10}$